INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

LIGHT, MARKER, GROUND-OBSTRUCTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a self-contained, battery-operated, steady or flashing, ground-obstruction marker light.

1.2 Classification. Marker lights shall be of the following battery types, color filter dome sizes, and carrying bag options, as specified (see 6.2).

1.2.1 Batteries. The type of battery shall be designated as follows:

Type
L - One lantern battery
C - Four "C" batteries

1.2.2 Color filter domes. The color filter dome sizes shall be designated as follows:

Size
1 - Small color filter domes in accordance with Figure 3
2 - Large color filter domes in accordance with Figure 4

1.2.3 Carrying bag options. The option of a carrying bag shall be designated as follows:

Class
Y - Carrying bag included
N - Carrying bag not included

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.3 Part or identifying number (PIN). A specification-based PIN shall be included to identify the battery type, color filter dome size, and carrying bag option for marker lights (see 6.2 and 6.8).

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

L-P-1183	- Plastic Molding Material, Acrylonitrile-Butadine-Styrene (ABS), Rigid.
V-T-285	- Thread, Polyester.
W-B-101	- Battery, Dry
WW-C-440	- Clamps, Hose. (Low Pressure).
CCC-C-428	- Cloth, Duck, Cotton: Fire, Water, Weather, and Mildew Resistant.
CCC-D-950	- Dyeing and Aftertreating Process for Cotton Cloths.

MILITARY

MIL-B-18/34	- Battery, Dry, BA-200/U.
MIL-V-173	- Varnish, Moisture-and-Fungus Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-501	- Pins, Tent, Metal.
MIL-C-7989	- Cover, Light-Transmitting, for Aeronautical Light, General Specification for.
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Provisional Items (Repair Parts): Packaging and Packing of.
MIL-C-25050	- Colors, Aeronautical Lights and Lighting Equipment, General Requirements for.
MIL-C-43307	- Cord, Nylon, Solid Braid, General Purpose.
MIL-C-43375	- Cloth, Duck, Nylon, 12.5 Ounce.
MIL-B-49030/3	- Battery, Dry, BA-3042/U.

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STANDARDS

FEDERAL

FED-STD-595	- Colors.
FED-STD-751	- Stitches, Seams, and Stitching.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	- Dissimilar Metals.
MS27977	- Fasteners, Snap, Style 1 (Large Curtain Type).

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications.. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 209	- Aluminum and Aluminum-Alloy Sheet and Plate.
ASTM D 256	- Impact Resistance of Plastics and Electrical Insulating Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

SOUTH COAST AIR QUALITY MANAGEMENT CONTROL DISTRICT

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management Control District, 9150 East Flair Drive, El Monte, CA 91731.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The marker light (hereinafter referred to as "light") shall consist of a light assembly, a ballast bag and four anchor pins (see 6.5 and figure 1).

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3 in accordance with 4.3).

3.3 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor, shall be of the lightest practicable weight suitable for the purpose intended, and shall be subject to all provisions of this specification. The material shall be free from all defects and imperfections that may affect the quality and reliability of the finished product. Lights fabricated of materials other than those specified herein, and any design changes required due to the use of such materials, shall be subject to approval of the contracting officer.

3.3.1 Material deterioration prevention and control. The light shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the light may be exposed. Plastics inherently susceptible to deterioration in ultraviolet light shall be treated (i.e., ultraviolet stabilized) for the effects of ultraviolet light exposure, except for plastics whose functions would be adversely affected by such treatment. When specified (see 6.2), a certification of ultraviolet treatment to plastic components shall be provided.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and subcomponents, and shall make information available upon request to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin

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raw materials. The components, pieces and parts incorporated in the light may be newly fabricated from recovered materials to the maximum extent practicable, provided the light produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the light.

3.4 Construction. The light shall be constructed so that parts will not work loose and replacement of the battery, dome, deflector, and lamp can be made by personnel without the use of special tools.

3.5 Operation. With the light switch (see 3.6.5) in the "ON" position, the light shall emit a steady glow. With the switch in the "OFF" position, the light shall be completely extinguished. With the switch in the "FLASH" position, the light shall flash in accordance with the flashing frequency and on time as specified in 3.6.6.

3.5.1 Operational performance. When tested as specified in 4.5.2.2, the light shall operate as specified in 3.5 without damage (see 6.4.1).

3.5.2 Impact. When tested as specified in 4.5.2.3, the light shall withstand a 6-foot drop without damage.

3.5.3 Rain resistance. When tested as specified in 4.5.2.4, the light shall operate in the "FLASH" position as specified in 3.5 without damage and without any noticeable variation in light output. There shall be no noticeable evidence of water penetration inside the canister or dome.

3.5.4 Low-temperature and high-temperature operation. When tested as specified in 4.5.2.5, the light shall operate as specified in 3.5 without damage between 32 °F (0 °C) and 131 °F (55 °C), and the flasher cycle rate shall not change more than ±10 percent between 32 °F (0 °C) and 131 °F (55 °C).

3.5.5 Endurance (operational reliability). When tested as specified in 4.5.2.6, the light shall operate in the "FLASH" position as specified in 3.5 without damage for a period of 50 hours (except for change of lamp) at 90 °F (32.2 °C).

3.6 Light assembly. The light assembly (see 6.5 and figure 2) shall consist of a cylindrical canister, deflector, clear dome, lamp, three-way switch, flasher, and such wiring and fasteners required to make a complete operational assembly when operated with a 6-volt carbon-zinc battery conforming to MIL-B-18/34, or four 1.5-volt alkaline "C" cells conforming to MIL-B-49030/3, or a 6-volt alkaline lantern battery,, or four 1.5-volt carbon-zinc "C" cells conforming to W-B-101. Batteries shall not be furnished with the light. A battery holder of commercial grade plastic to accommodate four "C" cells shall be furnished, on designs requiring a 6-volt lantern battery.

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3.6.1 Canister. The canister, including any canister cap, shall be made of high-impact plastic conforming to L-P-1183, types II through VI; or a high-impact plastic with a minimum Izod impact strength of 1.5 foot pounds/inch of notch (0.25-inch (0.64-cm) bar) at 73.4 ± 3.6 °F (23 ± 2 °C), when tested in accordance with method A of ASTM D 256. The canister shall contain the battery and electrical circuitry, and shall be compatible with the deflector and dome. A guard on the exterior of the canister shall be provided for the protection of the three-way switch. When color filter domes are furnished in accordance with figure 3 (see 3.6.3), there shall be space in the canister to store these domes without interference to the operation of the light.

3.6.2 Deflector. The deflector shall be either: (1) plastic with a nickel-chrome finish; (2) cold-rolled steel with a nickel-chrome finish; or (3) aluminum conforming to ASTM B 209, alloy 3003 with a polished anodic oxide finish. The deflector shall contain the light socket for the lamp (see 6.5 and figure 2).

3.6.3 Domes. The clear dome shall conform to figure 4 and shall be made of nondiffusing clear plastic conforming to MIL-C-7989, class D. An opaque dome shall be furnished as an accessory and shall conform to figure 4 and be made of plastic conforming to MIL-C-7989. Red, green, blue, and amber filter domes shall be furnished as accessories and shall conform to either figure 3 or figure 4. The chromaticity and transmittance of the color filter domes shall be as specified in MIL-C-25050, type I. When color filter domes are furnished in accordance with figure 3, they shall be stored within the canister (see 3.6.1), and when stored, shall not interfere with the operation of the light. The color filter domes conforming to figure 3 shall be made to fit over the lamp and maintain their position over the lamp while the light is inverted and shook (see 4.5.2.7). Also, when tested as specified in 4.5.2.5, the blue filter dome conforming to figure 3 shall not soften or deform when installed over the lamp and when the light is operated in the steady "ON" mode for 6 hours at an ambient air temperature of 131 degrees F (55 degrees C).

3.6.4 Lamp. The lamp shall be a commercial-type lamp, manufacturer's trade number 503. Two spare lamps shall be provided with each light (see 6.5 and figure 2).

3.6.5 Switch. An environmentally-sealed, externally-operated toggle switch shall be furnished. The switch shall provide "ON", "OFF", and "FLASH" positions. The "ON" position shall be labeled "ON", the "OFF" position shall be labeled "OFF", and the "FLASH" position shall be labeled "FLASH". The label characters shall be 0.18 inches tall and of a color contrasting with the background. A neoprene or silicone rubber boot with a nickel-plated brass mounting nut shall be provided to fit over the switch and prevent moisture from entering the switch and canister.

3.6.6 Flasher. The flasher or flasher circuit shall provide a flashing

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frequency of 60 cycles ±10 cycles per minute, and the "ON" time for the lamp shall be not less than 0.3-second duration for each flash. Cycle rate shall not change more than ±10 percent at ambient temperatures between 32 °F (0 °C) and 131 °F (55 °C).

3.6.7 Waterproofing gaskets. Gaskets used to seal the light from moisture shall be affixed to one of the components (see 6.5 and figure 2).

3.7 Ballast bag. The ballast bag (see 6.5 and figure 5) shall be made of either nylon duck cloth conforming to MIL-C-43375, class 2 or number 4 hard texture cotton duck cloth conforming to CCC-C-428, class I. The cotton duck cloth shall be treated with either mildew inhibitor (a) or (e) of CCC-D-950. Thread conforming to V-T-285, type I, class I, shall be used for all stitches. Stitches shall conform to FED-STD-751, type 301 with 5 to 7 stitches per inch (2.54 cm). The ballast bag shall be provided with one 12-inch (30.48 cm) opening ± 0.5 inch (1.27 cm) for filling and emptying the bag. Each opening shall be provided with three snap fasteners conforming to MS27977-2N for the socket, long prong; MS27977-3N for the clinch plate; MS27977-3N for the stud, eyelet base; and MS27977-10N for the washer. All fasteners shall be corrosion resistant with a flat black finish. The ballast bag shall be marked "FILL WITH BALLAST BEFORE USING", in black letters not less that 0.375 inch (0.95 cm) high.

3.7.1 Holddown lines. Two nylon lines 0.25 inch (0.64 cm) in diameter conforming to MIL-C-43307 shall be stitched diagonally from corner to corner to the bottom of the ballast bag. The nylon lines shall not overlap where the lines intersect at the bottom center of the ballast bag. Each end of the holddown lines shall be provided with a 2-inch (5.08 cm) diameter stitched loop. The four loops shall project 6 inches (15.24 cm) ± 0.5 inch (1.27 cm) from the bottom corners of the ballast bag.

3.7.2 Collar. On designs requiring a collar, the collar shall be provided with a clamp conforming to WW-C-440, type F, corrosion-resisting steel, 4.5-inch (11.43 cm) nominal clamp size with thumb screw adjusting mechanism. The clamp shall pass through a minimum of two stitched loops (see 6.5 and figure 1).

3.8 Anchor pins. Four anchor pins, each 9 inches (22.86 cm) in length, shall be provided with each light. Pins shall conform to MIL-P-501, type I, aluminum (except that the color shall be as specified in 3.14) or L-P-1183, type VI, plastic, similar in construction to the type I aluminum pin.

3.9 Carrying bag. When specified (see 1.2 and 6.2), a nylon or cotton duck cloth carrying bag shall be provided for carrying a set of eight lights. The nylon duck cloth shall conform to MIL-C-43375, class 2. The cotton duck cloth shall be number 4 hard texture cotton duck cloth conforming to CCC-C-428, class 1. The cotton duck cloth shall be treated with either mildew inhibitor (a) or (e) of CCC-D-950. Thread conforming to V-T-285, type I, class I, shall be used

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for all stitches. Stitches shall conform to FED-STD-751, type 301 with 5 to 7 stitches per inch (2.54 cm). The carrying bag shall be provided with two carrying handles or straps. An opening shall be provided for installing the set of eight marker lights. The opening shall be provided with snap fasteners conforming to MS27977-2N for the socket; MS27977-3N for the clinch plate; MS27977-8N for the stud; and MS27977-10-N for the washer. All fasteners shall have a flat black finish. The carrying bag shall be marked "CARRYING BAG FOR MARKER LIGHTS" in black letters not less than 0.500 inch (1.27 cm) high.

3.10 Interchangeability. All parts having the same contractor's part number shall be directly and completely interchangeable with each other with respect to installation and performance.

3.11 Identification marking. The light shall be identified in accordance with MIL-STD-130.

3.12 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I or II, as applicable (see 3.12.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

a. Components or circuit elements that are inherently fungus-and-moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1-mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.12.1 Composition. Composition II shall be used only in the case when local air pollution regulations in the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Rule 442, South Coast Air Quality Management District of California.

3.13 Dimensions. The dimensions of the light, with 20 pounds (9 kg) of sand as ballast, shall not exceed the following:

Length - 12.5 inches (31.75 cm)
Width - 12.5 inches (31.75 cm)
Height - 11.5 inches (29.21 cm)

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3.14 Color. The color of the canister, ballast bag, thread, holddown lines, anchor pins and carrying bag shall be aviation yellow number 13538, number 23538 or number 33538 as specified in FED-STD-595.

3.15 Workmanship. The light, including all parts and accessories shall be fabricated and finished to ensure freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of parts, and tightness of assembly components.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

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4.3 First article inspection.

4.3.1 Examination. The light shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The light shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, inspection level S-2.

4.4.2 Examination.

4.4.2.1 Samples. Samples selected in accordance with 4.4.1 shall be examined for the major and minor characteristics for possible defects as specified in 4.5.1. Acceptable quality level (AQL) shall be 1.0 percent defective for major defects and 2.5 percent defective for minor defects.

4.4.3 Tests.

4.4.3.1 Individual. Each light shall be tested as specified in 4.5.2.1 and 4.5.2.2. Failure of the test shall be cause for rejection.

4.4.3.2 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.2, 4.5.2.3, 4.5.2.4, 4.5.2.5, 4.5.2.6, and 4.5.2.7. Tests shall be conducted in the sequence shown. AQL shall be 1.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The sample lights shall be examined as specified in 4.4.2.1 for the following defects:

Major

101. Parts or components missing, or out of place, damaged or do not function (see 3.1).
102. Assembly incomplete or not correct (see 3.1).
103. Materials not as specified (see 3.3).
104. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments, and when required, certification of ultraviolet treatment to plastic components not provided (see 3.3.1).

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105. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.1.1).
106. Contractor does not have documentation available for identification of material, material finishes, or treatment (see 3.3.1.2).
107. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the light (see 3.3.2).
108. Construction not as specified (see 3.4).
109. Light assembly not as specified (see 3.6).
110. Canister not as specified (see 3.6.1).
111. Deflector not as specified (see 3.6.2).
112. Domes not as specified (see 3.6.3).
113. Lamp not as specified (see 3.6.4).
114. Switch not as specified (see 3.6.5).
115. Gaskets not as specified (see 3.6.7).
116. Ballast bag not as specified (see 3.7).
117. Holddown lines not as specified (see 3.7.1).
118. When required, collar not as specified (see 3.7.2).
119. Anchor pins missing or not as specified (see 3.8).
120. When required, carrying bag not as specified (see 3.9).
121. Parts not interchangeable (see 3.10).
122. Fungus and moisture resistance not as specified (see 3.12).
123. Composition not as specified (see 3.12.1).
124. Dimensions not as specified (see 3.13).
125. Color not as specified (see 3.14).
126. Workmanship not as specified (see 3.15).

Minor

201. Identification marking missing or not as specified (see 3.11).

4.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified, tests shall be conducted at standard ambient temperature as defined in MIL-STD-810. Prior to each test, a new 6-volt battery or four new 1.5-volt "C" cell batteries, as applicable, shall be installed in the light and the light shall be operated in each mode ("ON", "OFF", and "FLASH") to determine that the light meets the requirements of 3.5.

4.5.2.2 Operating. Operate the light in each mode ("ON", "OFF", and "FLASH") five times. Failure to operate in each mode five times without damage shall constitute failure of this test.

4.5.2.3 Impact. Fill the ballast bag with 20 pounds (9 kg) of sand. Drop the light from a height of 6 feet (1.83 m) upon a hard, rigid surface. Then operate the light in each mode. Evidence of spare lamps that have fallen out of the friction sockets, any damage or inability of the light to operate in

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each mode as specified in 3.5 after the drop shall constitute failure of this test.

4.5.2.4 Rain. Subject the light to a rain test in accordance with MIL-STD-810, method 506.2, procedure III. During the test, the light shall be operating in the "FLASH" mode. Nonconformance to 3.5.3 shall constitute failure of this test.

4.5.2.5 Low-temperature and high-temperature operation. Cold soak the light, less battery, for 4 hours at 32 °F (0 °C). Install the battery and operate the light for 15 minutes in the "FLASH" mode. At the end of 15 minutes, return the light to ambient temperature and repeat the test specified in 4.5.2.2. Then hot soak the light, less battery, for 4 hours at 131 °F (55 °C). Install the battery and operate the light for 15 minutes in the "FLASH" mode. When a blue filter in accordance with figure 3 is furnished (see 3.6.3), install this filter in place of the clear dome, install a new battery, and operate the light in the steady "ON" mode for six hours at 131 °F (55 °C). Return the light (with either the clear dome or blue filter, as applicable) to ambient temperature and repeat the test specified in 4.5.2.2. Nonconformance to 3.5.4 and 3.6.3, as applicable, shall constitute failure of this test.

4.5.2.6 Endurance. Install a new 6-volt battery or four new 1.5-volt "C" cell batteries, as applicable, in the light. Operate the light continuously in the "FLASH" mode for not less than 50 hours in an ambient temperature of 90 °F (32.2 °C). Nonconformance to 3.5.5 shall constitute failure of this test.

4.5.2.7 Color filter dome adhesion. When applicable (see 3.6.3), invert the light assembly and shake vigorously. Nonconformance to 3.6.3 shall constitute failure of this test.

4.6 Inspection of packaging. The preservation, packing, and marking shall be examined to determine compliance with the quality assurance provisions of MIL-E-17555 and any applicable documents referenced therein.

5 PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.2). Each complete light, with carrying bag when specified, shall be preserved in accordance with the level A or C requirements of MIL-E-17555.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2). Lights, preserved as specified in 5.1, shall be packed in accordance with the level A, B, or C requirements of MIL-E-17555.

5.3 Marking. In addition to any special marking specified in the contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The light covered by this specification is intended for use as an emergency marker light for applications requiring portability and short-range visibility from the air, such as marking closed runways, helicopter operations, soft spots, or runway lighting on aircraft landing strips.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Battery type, color filter dome size, and carrying bag option required (see 1.2).
c. Part or identifying number (PIN) required (see 1.3).
d. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
e. When a first article is not required (see 3.2).
f. When a certification of ultraviolet treatment to plastic components shall be provided (see 3.3.1).
g. Level of preservation and packing required (see 5.1 and 5.2).
h. Special marking required (see 5.3).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions. The following definitions should apply throughout this specification.

6.4.1 Damage. Damage is deformation, loosening, separation of the canister from the ballast bag, breakage, change of fit of any component or part, corrosion, or any condition resulting in malfunction of the marker light or any component or part.

6.5 Information figures. Figures 1, 2 and 5 show a type of marker light which has been found acceptable; however, the figures are included for

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illustration only and are not intended to preclude the furnishing of other marker lights that conform to this specification.

6.6 Dimensional figures. Figures 3 and 4 show dimensions for the applicable dome that must be achieved (see 3.6.3).

6.7 Subject term (key word) listing.

Ballast bag
Blinking light
Emergency light
Flasher
Flashing light
Light
Light assembly
Marker
Portable light
Strobe light

6.8 Part or identifying number (PIN). The PIN to be used for marker lights acquired to this specification is created as follows:

M	52543	X	X	X
Military specification prefix	Specification number	Battery type L or C (see 1.2.1)	Filter dome size 1 or 2 (see 1.2.2)	Carrying bag option Class Y or N (see 1.2.3)

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - AS

Review activity:
DLA - GS

User activity:
Army - AV

Preparing activity:
Army - ME

Project 6230-0300

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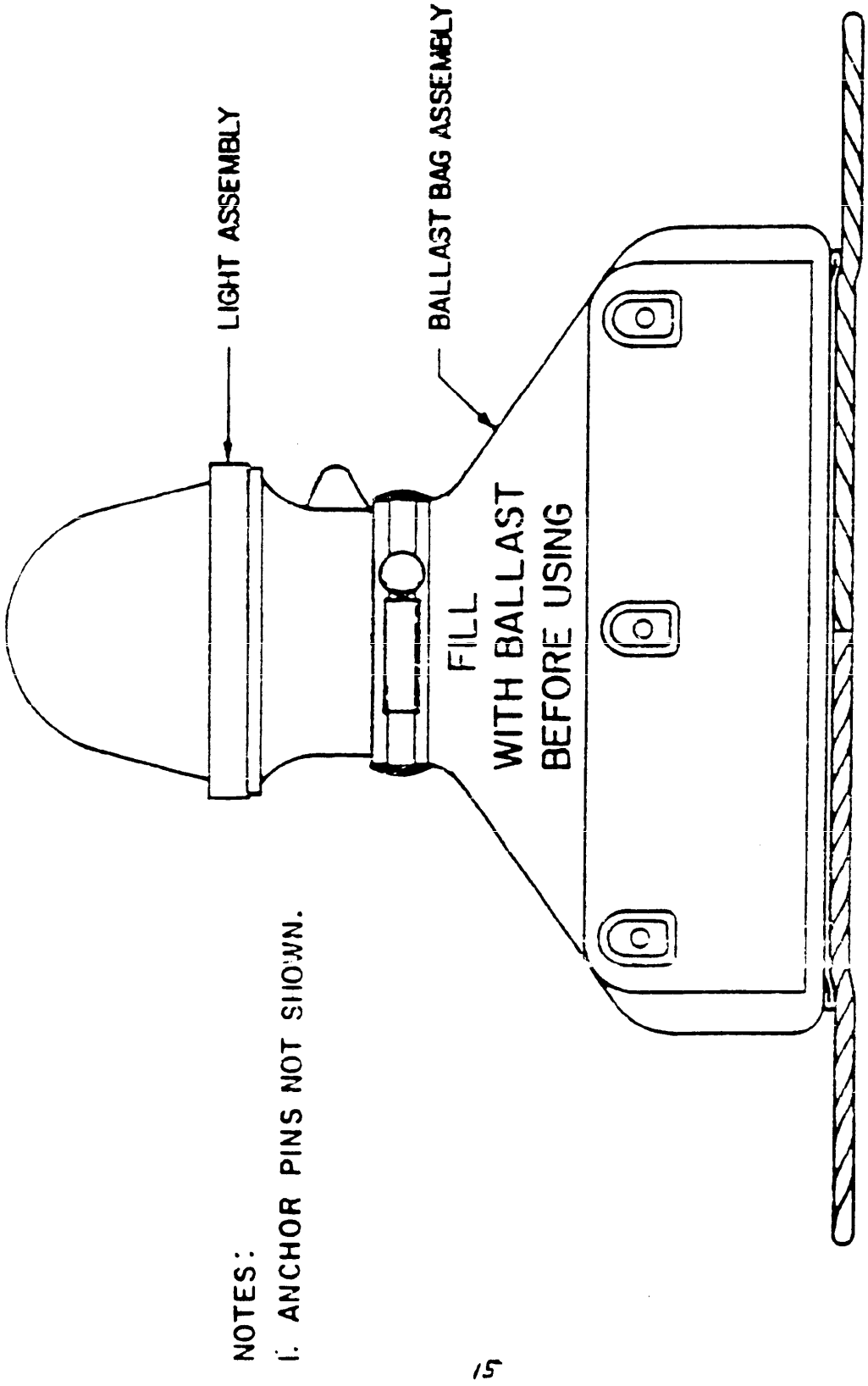
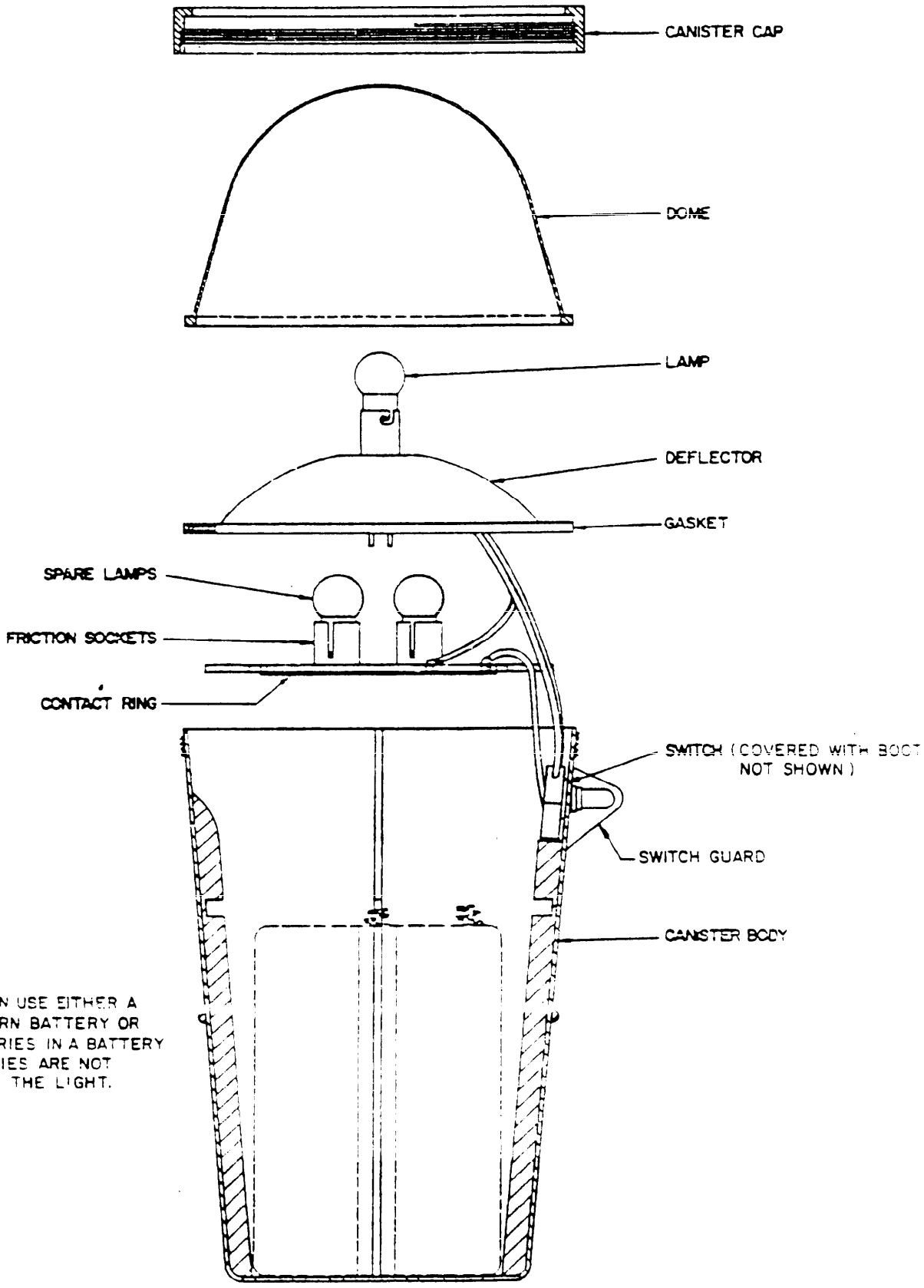


FIGURE 1. Light, marker, ground obstruction.

XI620B

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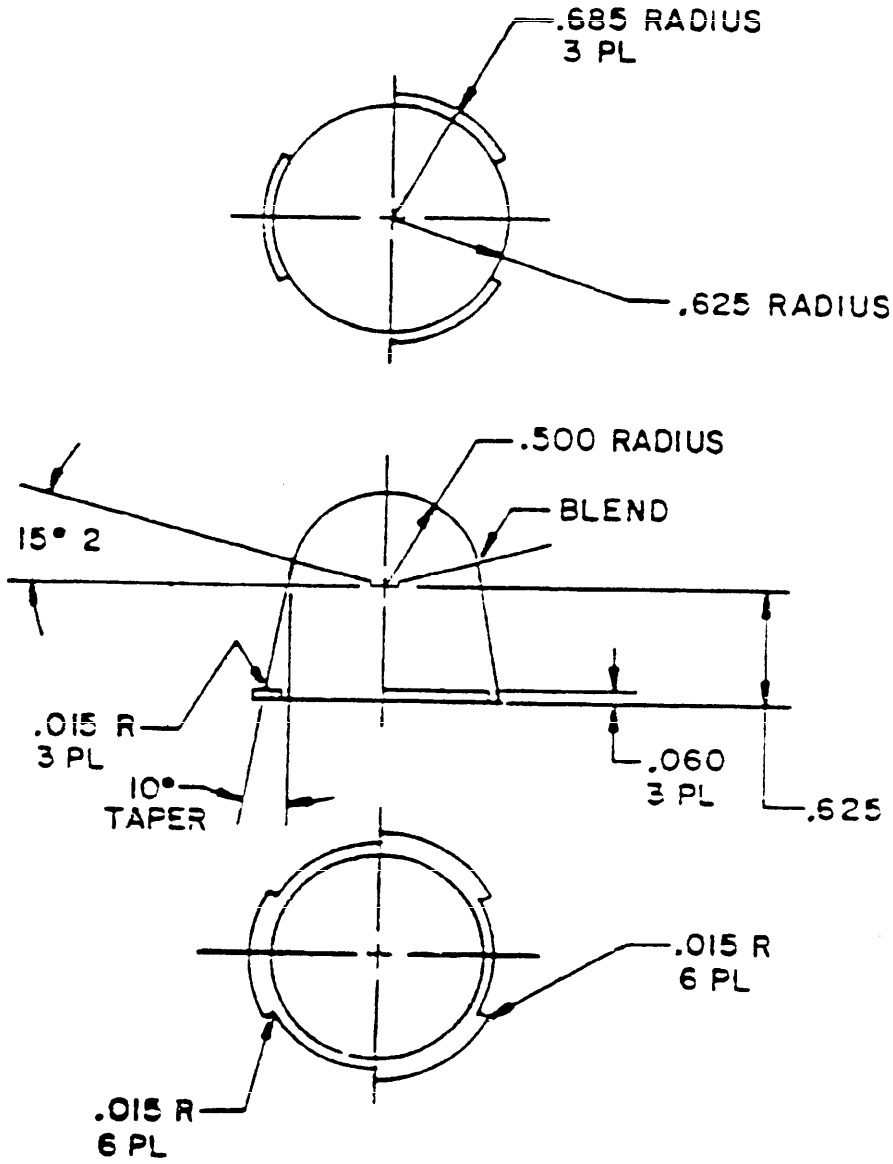
NOTES:

1. LIGHT SHOWN CAN USE EITHER A SIX-VOLT LANTERN BATTERY OR FOUR "D" BATTERIES IN A BATTERY HOLDER. BATTERIES ARE NOT FURNISHED WITH THE LIGHT.

FIGURE 2. Light assembly.

X162I A

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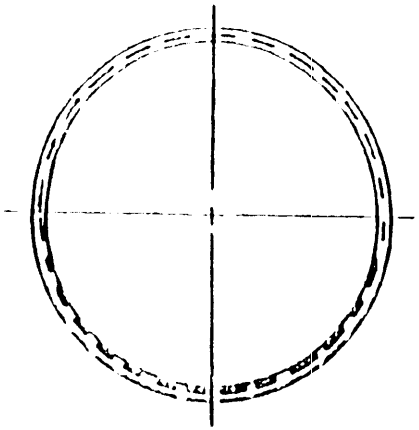
NOTES:

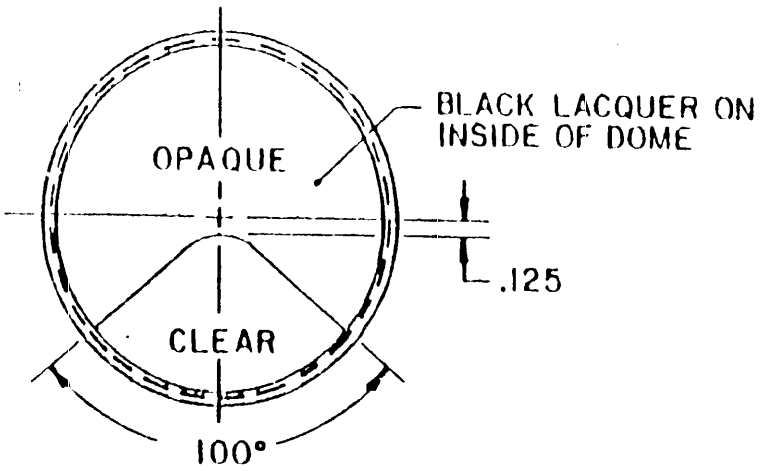
1. GREEN, BLUE, AMBER, AND RED FILTER DOMES.
2. ALL DIMENSIONS ARE IN INCHES.
3. NOMINAL WALL THICKNESS .060.
4. TOLERANCE ±.005.

FIGURE 3. Molded small color filter domes.

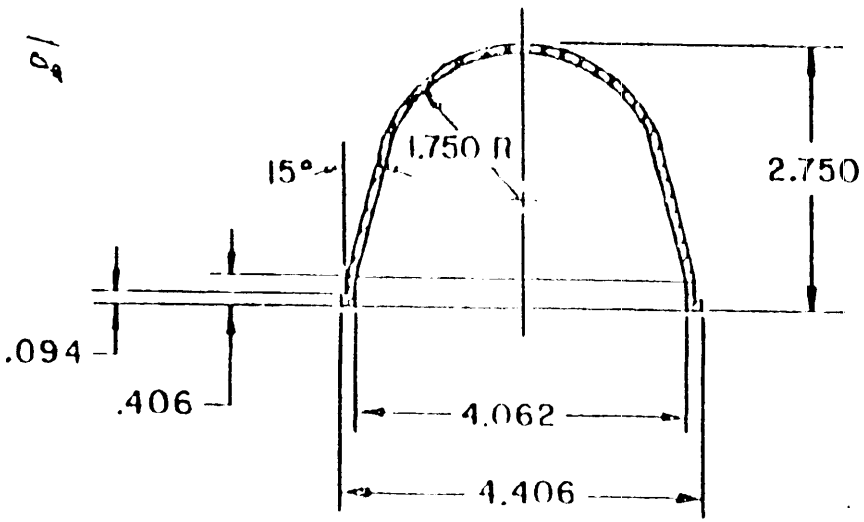
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MARKER, GROUND OBSTRUCTION LIGHT





OPAQUE DOME



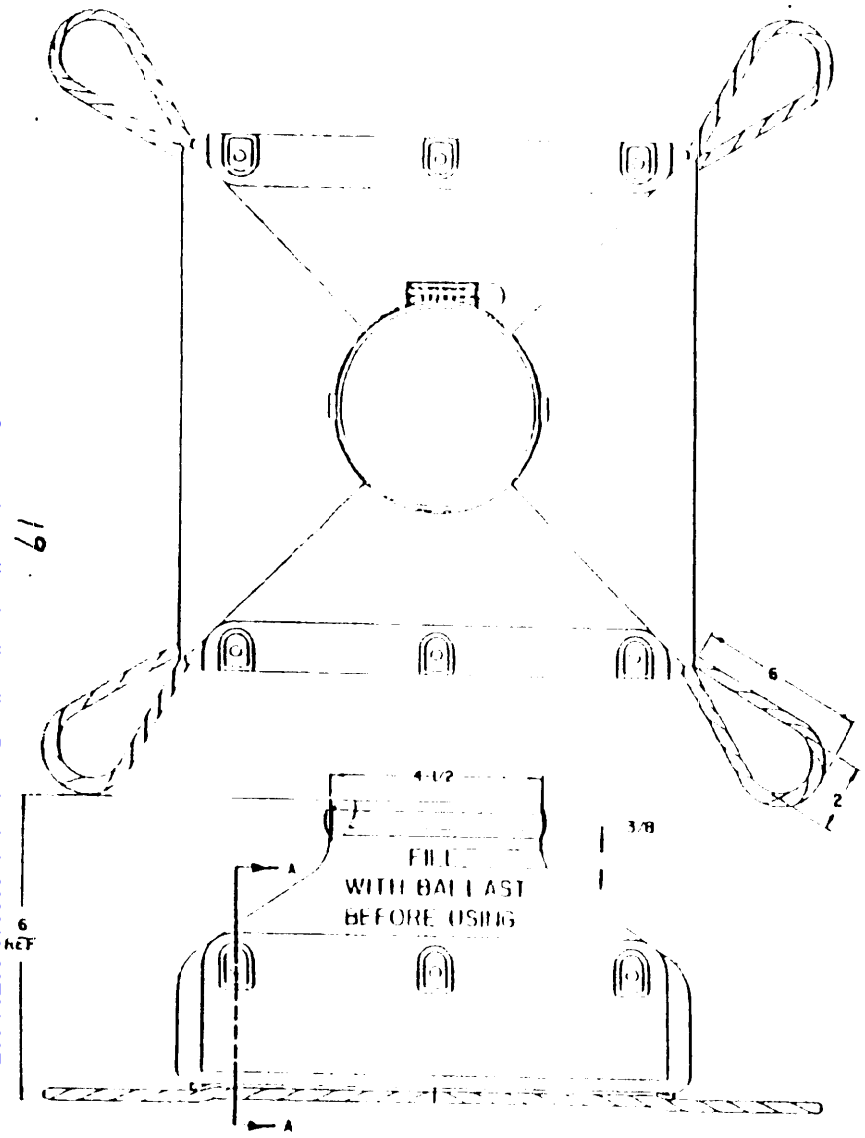
CLEAR, GREEN, BLUE, AMBER,
AND RED DOMES

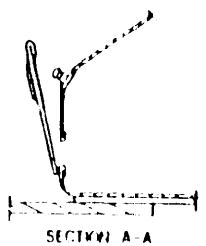
NOTES:

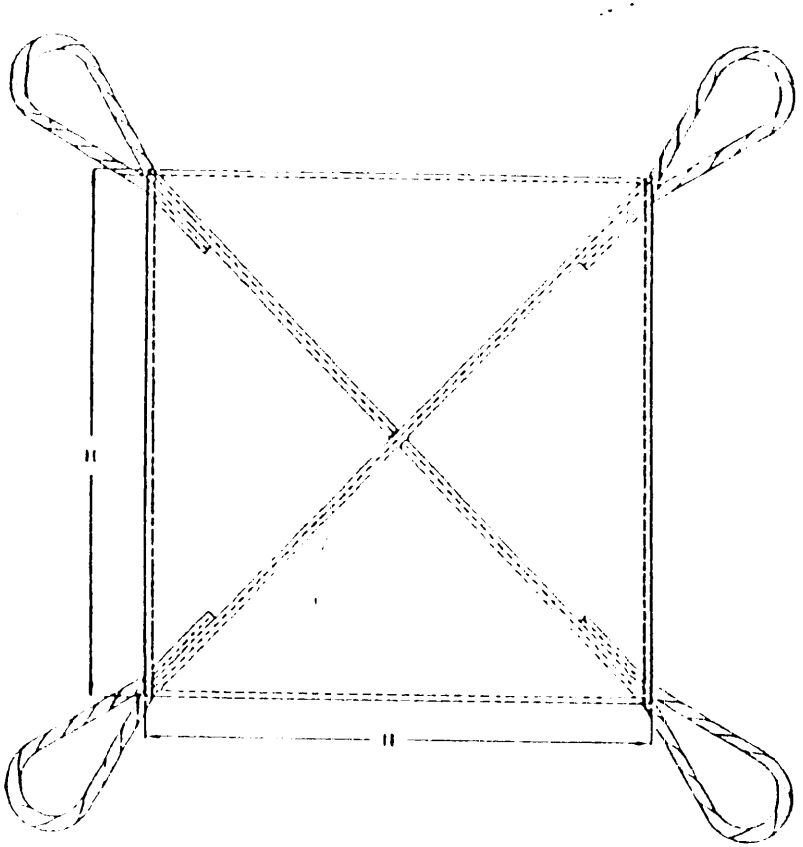
1. ALL DIMENSIONS IN INCHES.
2. UNSPECIFIED TOLERANCES ARE:
   ANGLES ±1°
   DECIMALS ±.031.
3. COLOR FILTER DOMES, OPAQUE DOME, AND CLEAR DOME HAVE SAME DIMENSIONS.

FIGURE 4. Molded large clear, opaque and color filter domes.

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NOTE:
ALL DIMENSIONS IN INCHES

MIL-I-52543D

FIGURE 5. Bag, ballast assembly.

[A 162-1A]

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-52543D	Light, Marker, Ground-Obstruction

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYYYMMDD)*

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